Multi-Scale Predictability of High-Impact Weather in the Battlespace Environment

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1. Introduction

The accurate prediction of the coastal atmosphere-ocean environment is critical to the success of US Navy operations. Operations, such as strike warfare, special operations, carrier flight operations, mission planning and rehearsal, and sorties, may be adversely affected by the high-frequency spatial and temporal variations of environmental conditions that occur in the battlespace environment. The presence of horizontal and vertical wind shears, precipitation, clouds, ducting conditions, pronounced temperature and/or moisture gradients, and aerosols are among the conditions that may adversely impact Navy operations. Because of the chaotic nature of the atmosphere-ocean system, and its sensitive dependence on initial conditions, there will always be uncertainty associated with forecasts of these phenomena. Due to inevitable uncertainties in the initial state, forecasts are most appropriately viewed in a probabilistic sense; that is, forecasts should provide probabilities of the occurrence of specific events, as well as estimates of forecast skill (e.g., Leith, 1974). The only computationally feasible approach to this problem is to perform an ensemble of forecasts from equally plausible initial states. Due to the high dimension of the system, a large number of ensembles are necessary to accurately forecast the probability that a specific high-impact event will occur. These high-impact events can be extreme events, such as strong windstorms that generate aerosols and impact electro-optical (EO) conditions, or relatively weak forcing events such as stratus clouds that may impact flight operations and electro-magnetic (EM) ducting conditions.

While progress has been made in ensemble forecasting, many significant questions remain unanswered. There is still considerable uncertainty and disagreement on the most effective way to perturb the initial conditions for both global scale and mesoscale models. Model errors and uncertainties should also be included in ensemble configuration, yet much of that work remains in its infancy. Inclusion of model uncertainties will become particularly important for tropical phenomena such as tropical cyclones. Here we report on recent advances in ensemble design facilitated by the HPC program, including research on the impact of stochastic model perturbations and comparison of different methods for perturbing initial conditions.

2. Objective

The objective of this project is to determine the predictability and define the methodologies for effective ensemble prediction of high-impact weather in the battle-space environment. To accomplish this goal, we test and evaluate methodologies for the generation of initial condition perturbations in ensemble forecasts. We also test and evaluate methodologies for the inclusion of model error in ensemble design, specifically testing the impact of stochastic perturbations to the model tendencies produced by the parameterization of deep convection. The first parameters that we examine here are tropical cyclone track forecasts, and upper and lower tropospheric wind speed.

3. Methodology

The research we focus on for this report concerns experiments with the Navy Operational Global Atmospheric Prediction System (NOGAPS). First, a description of NOGAPS is given, followed by a description of the stochastic perturbations introduced in the ensemble to represent model uncertainty, followed by a description of the initial perturbation techniques that represent initial condition uncertainty.

3.1. Model Description

The NOGAPS atmospheric model is a global spectral model (Hogan and Rosmond, 1991) with a hybrid-sigma coordinate in the vertical. The prognostic variables are vorticity, divergence, virtual potential temperature,

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specific humidity and terrain pressure. The parameterizations of physical processes include deep convective precipitation (Emanuel and Zivkovic-Rothman, 1999); shallow cumulus mixing (Tiedtke, 1984); convective and stratiform clouds (Slingo, 1987); boundary layer vertical mixing (Louis et al. 1982); surface flux parameterization (Louis 1979); gravity wave drag (Palmer et al., 1986); and the Harshvardhan et al. (1987), radiation scheme. The NOGAPS atmospheric model can be run at a variety of spectral resolutions.

3.2. Stochastic Perturbations

Following Teixeira and Reynolds (2006), we develop a methodology to utilize physical parameterizations in a stochastic manner in the context of ensemble prediction, based on the knowledge of the variance provided by the parameterization. Assume that the value of a generic variable after being updated by a certain physical parameterization (moist convection in the present study) can be written as:

$$\phi_{conv}^{stoch} = \overline{\phi}_{conv} + \varepsilon \,, \tag{1}$$

where $\overline{\phi}_{conv}$ is the mean value of the variable after convection, ϕ_{conv}^{stoch} is the stochastic value of the same variable after convection and ε is a normally distributed stochastic variable with mean $\mu(\varepsilon) = 0$ and standard deviation $\sigma(\varepsilon) = \sigma_{\phi,\ conv}$, where $\sigma_{\phi,\ conv}$ is the standard deviation of the variable due to moist convective processes. In practice, this perturbation is created as such:

$$\left(\frac{\partial \phi}{\partial t}\right)_{conv}^{stoch} = \left(1 + \eta\beta\right)\left(\frac{\partial \overline{\phi}}{\partial t}\right)_{conv} . \tag{2}$$

where β is a constant of proportionality and η is a normally distributed stochastic variable with mean $\mu(\eta) = 0$ and standard deviation $\sigma(\eta) = 1$. Ensemble experiments are first performed where only model uncertainty in the form of stochastic perturbations is considered. The perturbations are only added to the tendencies of the model parameterization of deep convection. These are followed by experiments that combine these stochastic perturbations with perturbations in the initial conditions, as described below.

3.3. Initial Perturbations

A new way of perturbing the initial conditions in the ensemble are compared with the operational ensemble perturbation method. The current operational technique is based on the bred vector (BV) method first introduced by Toth and Kalnay (1993). In this technique, the initial perturbations are based on short-term evolved ensemble perturbations that have been rescaled to reflect spatial differences in the global observing system. This operational technique is compared with a new technique called the ensemble transform (ET) method, first proposed by Bishop and Toth (1999). In the ET method, the initial perturbations are produced using evolved forecast perturbations, as in the BV method, but in the ET technique, the transformation is performed such that the initial perturbations are consistent with the analysis error variance estimates produced by the operational 3-dimensional (3-D) variational data assimilation scheme, NAVDAS (Daley and Baker, 2001). In some experiments, the ET and BV ensembles are also compared to a third method, the analysis error simulator (AES) that has also been under development at NRL. As with the ET, AES uses the NAVDAS analysis error variance estimates, but, unlike the ET, it is a non-cycling scheme.

4. Results

4.1. Impact of Stochastic Perturbations on Ensemble Spread

Figure 1 shows the ensemble perturbation standard deviation of the upper tropospheric zonal wind induced by stochastic perturbations only, for different forecast times. Because the perturbations are made only to tendencies produced by the parameterization of sub-grid-scale deep convection, the 24-h ensemble spread is concentrated in regions where deep convection is frequent, most notably the tropics. However, as time progresses, significant perturbation growth also occurs in the mid latitudes, both through local deep convection processes, and through influences of the perturbed tropical fields on the mid latitudes.

4.2. Impact of Stochastic Perturbations on Tropical Cyclone Track Forecasts

The significant impact in the tropics is also noted in an experiment to examine the impact on tropical cyclone tracks. The stochastic perturbations significantly increase the ensemble spread in tropical cyclone track forecasts. We show an example for ensemble forecasts for Hurricane Charley from 10 August 2004 (Figure 2). This is almost four days before landfall in southwest Florida. For the ensemble based on initial perturbations alone, all the ensemble forecasts predict that Charley will continue on a westerly path. When stochastic perturbations representing model error are added to the ensemble formulation, at least a few of the ensemble members capture the possibility that Charley will recurve to the north and strike south Florida.

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4.3. Comparison of the "Spread-Skill" Relationship for the Different Methods

The "spread-skill" relationship is one of the most important characteristics of an ensemble. Ensembles give reliability estimates of the skill of the forecast through the ensemble variance or "spread". Large (small) spread should be related to large (small) error, on average. In Figure 3, the spread skill relationship for the 10-m wind for the 72-h forecast is compared for the different methods. The two new methods (AES and ET) show better spread-skill relationships (closer to the diagonal line) then the operational ensemble. Including stochastic physics improves this relationship even further.

4.4. Ensemble Mean Error and Bias Removal

The ensemble archive provides an efficient way to calculate and remove the model bias from the current forecast. Figure 4 compares the single (control) forecast with the ensemble mean forecast, based on the ET method, with and without the bias removed, for the tropical 10-m winds, as a function of forecast time. The ensemble mean forecasts have lower error than the single forecasts. Removing the bias reduces the error from the control forecasts considerably (no bias removal). Adding stochastic physics reduces the forecast error further. A 96-h ensemble-mean forecast with the bias removed is as good as the 48-h single control forecast. This plot concisely illustrates: 1) the improvement of the ensemble mean over a single deterministic forecast, 2) the improvement achievable through bias correction, and 3) further improvement to the ensemble mean through inclusion of stochastic perturbations.

5. Summary and Significance to DoD

This research in ensemble design shows that new methods being developed for the Navy's ensemble forecasting system will bring considerable improvement over the current operational system. It is found that the inclusion of stochastic perturbations has a significant impact on ensemble skill in the tropics, especially on tropical cyclone track forecasts. New methods for perturbing the initial conditions provide superior ensemble performance in terms of ensemble mean error and spread-skill relationships when compared to the current operational system.

The Navy Operational Global Atmospheric and Prediction System (NOGAPS) is a critical, central component of the day-to-day operations of the Department of Defense (DoD). NOGAPS is a state-of-the-art, global numerical weather prediction system that provides global weather prediction products to the entire DoD community. NOGAPS is run four times daily at Fleet Numerical Meteorology and Oceanography Center (FNMOC) and provides global analyses and 7-day forecast fields. NOGAPS fields are used to drive a large suite of ocean, ice, and wave models, aircraft and ship routing software, and to provide boundary conditions for coupled ocean/atmosphere mesoscale prediction systems (COAMPS). The fields produced from these forecast models are used by the US Navy and Air Force throughout the world for weather guidance in support of real-time operations, mission planning, mission rehearsals, etc. COAMPS and NOGAPS are also used for tropical cyclone guidance at the Joint Typhoon Warning Center (JTWC) and are the backup NWP systems for the National Weather Service. This work will result in critical upgrades to the current forecasting capability of the DoD by providing reliable probabilities of the occurrence of specific high-impact weather events, and providing information as to the expected skill and reliability of a forecast. It will lead to significant improvements over the current 9-member FNMOC global ensemble forecasts and lead to mesoscale ensemble forecasting capabilities.

Our COAMPS and NOGAPS developmental strategies are consistent with the Commander, Naval Meteorology and Oceanography Center (CNMOC) Strategic Plan and will provide environmental support for Sea Power 21 and ForceNet. NRL-MRY is focused on ensuring that both NOGAPS and COAMPS will be an end-to-end system that integrates, tests, evaluates, and transitions to operations, all components necessary to meet this Strategic Plan. In particular, with the assistance obtained through this work, we are addressing the development of multi-scale probabilistic forecasts to meet the needs of the Navy.

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References

Bishop, C.H. and Z. Toth, "Ensemble transformation and adaptive observations." *J. Atmos. Sci.*, 56, pp. 1748–1765, 1999.

Emanuel, K.A. and M. Zivkovic-Rothman, "Development and evaluation of a convection scheme for use in climate models." *J. Atmos. Sci.*, 56, pp. 1766–1782, 1999.

Harshvardhan, R. Davies, D. Randall, and T. Corsetti, "A fast radiation parameterization for atmospheric circulation models." *J. Geophys. Res.*, 92, pp. 1009–1015, 1987.

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Hodur, R., "The Naval Research Laboratory's coupled ocean/atmosphere mesoscale prediction system (COAMPS)." *Monthly Weather Review*, 125, pp. 1414–1430, 1997.

Hogan, T.F. and T.E. Rosmond, "The description of the U.S. Navy Operational Global Atmospheric Prediction System's spectral forecast model." *Mon. Wea. Rev*., 119, pp. 1786–1815, 1991.

Leith, C.E., "Theoretical skill of Monte-Carlo forecasts." *Mon. Wea. Rev*., 102, pp. 409-418, 1974.

Louis, J.F., "A parametric model of vertical eddy fluxes in the atmosphere." *Bound.-Layer Meteor*., 17, pp. 187–202, 1979.

Louis, J.F., M. Tiedtke, and J.F. Geleyn, "A short history of the operational PBL parameterization at ECMWF." *Proc. ECMWF Workshop on Boundary-Layer Parameterization*, Reading, UK, ECMWF, pp. 59–79, 1982.

Palmer, T.N., G.J. Shutts, and R. Swinbank, "Alleviation of a systematic westerly bias in general circulation and numerical weather prediction models through an orographic gravity wave drag parameterization." *Quart. J. Roy. Meteor. Soc*., 112, pp. 1001–1039, 1986.

Slingo, J.M., "The development and verification of a cloud prediction scheme in the ECMWF model." *Quart. J. Roy. Meteor. Soc*., 113, pp. 899–927, 1987.

Teixeira, J. and C.A. Reynolds, "The stochastic nature of physical parameterizations in ensemble prediction: a stochastic convection approach." *J. Atmos. Sci*., 2006 (submitted).

Tiedtke, M., "The sensitivity of the time-scale flow to cumulus convection in the ECMWF model." *Proc. ECMWF Workshop on large-Scale Numerical Models*, Reading, UK, ECMWF, pp. 297–316, 1984.

Toth, Z, and E. Kalnay, "Ensemble forecasting at NMC: The generation of perturbations." *Bull. Am. Meteorol. Soc*., 74, pp. 2317–2330, 1993.

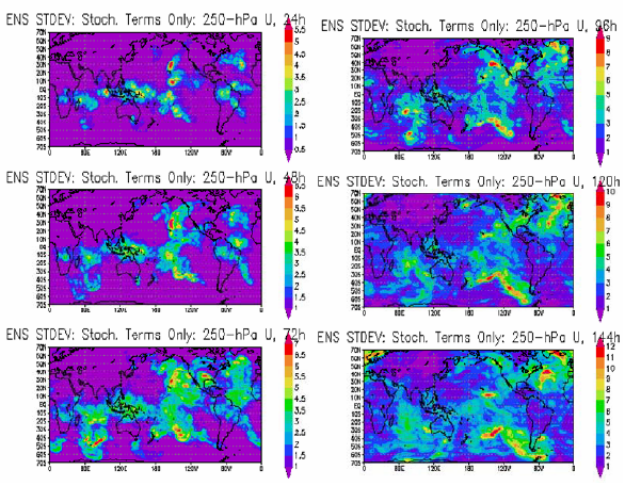

Figure 1. The ensemble standard deviation (m/s) of the 250-hPa zonal wind produced from an ensemble with stochastic perturbations only, for forecast times ranging from 24 to 144 hours

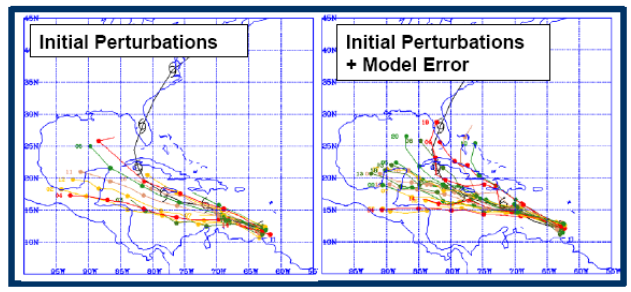

Figure 2. The ensemble storm tracks for Hurricane Charley from 10 August 2004 for ensembles with just initial condition perturbations on the left, and initial perturbations plus stochastic perturbations on the right. The observed track of Charley is denoted by the black track.

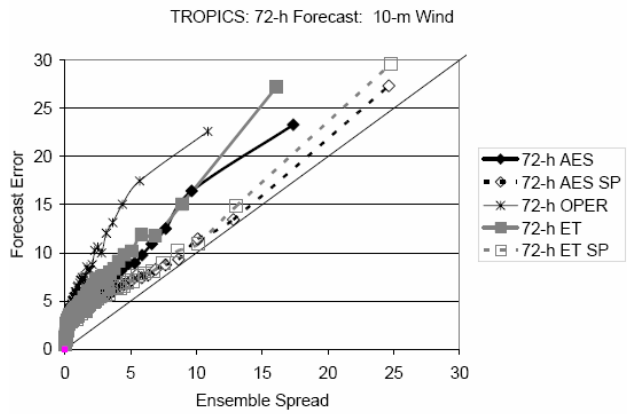

Figure 3. The relationship between Ensemble Spread and Forecast Error Variance for the different ensemble techniques, AES, ET and OPER (operational bred vectors) for the 10-meter zonal wind for 72-h forecasts, averaged for August-September 2005. SP denotes inclusion of stochastic perturbations.

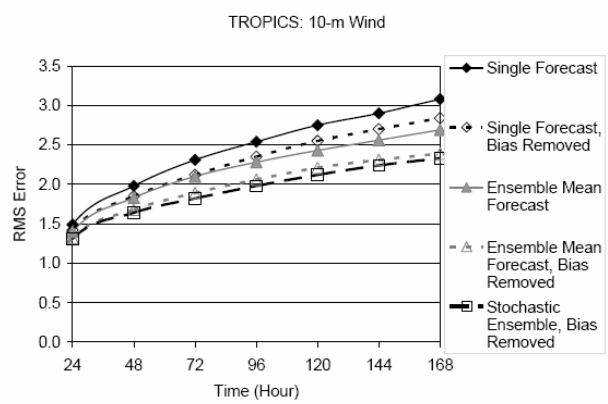

Figure 4. The root mean square error for the 10-meter zonal wind for 72-h forecasts for the tropics, averaged over August and September 2005. Different ensemble techniques denoted in key.

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